

Managing Parts Reuse/Replace Decisions with Dedicated Resource

Application Maintenance Site Management **Component Rebuild** Safety MARC Management

Managing Parts Reuse/Replace Decisions with Dedicated Resource	0
1.0 Introduction	1
2.0 Best Practice Description	1
3.0 Implementation Steps.....	2
4.0 Benefits	3
5.0 Resources Required.....	3
6.0 Supporting Attachments / References.....	3
7.0 Related Best Practices	3
8.0 Acknowledgements	3

DISCLAIMER: The information and potential benefits included in this document are based upon information provided by one or more Cat® dealers, and such dealer(s) opinion of “Best Practices”. Caterpillar makes no representation or warranty about the information contained in this document or the products referenced herein. Caterpillar welcomes additional “Best Practice” recommendations from our dealer network.

1.0 Introduction

Caterpillar has many guidelines on parts reuse criteria, and inspection / measurement procedures to determine reusability. These guidelines are primarily technical guidelines and do not address a “process” by which to determine parts reuse during component rebuilds.

Parts “reuse” or “replace” decisions during major component rebuild are one of the most important controllable cost drivers in the rebuilding process. Proper reuse decisions are vital. Rebuild cost increases when parts are replaced, that could have been reused. If the rebuild is a component exchange flat rate rebuild or MARC contract rebuild, this action directly reduces CRC margins. In addition to increased costs and reduced margins, poor rebuild/reuse decisions can result in component failure or shortened component life.

Often, decisions to replace or reuse a part will be made by virtually any mechanic in the shop. This decision often lacks consistency from one mechanic to another due to experience, training, and even a fear of risk.

This Best Practice discusses managing parts reuse decisions through a consistent process.

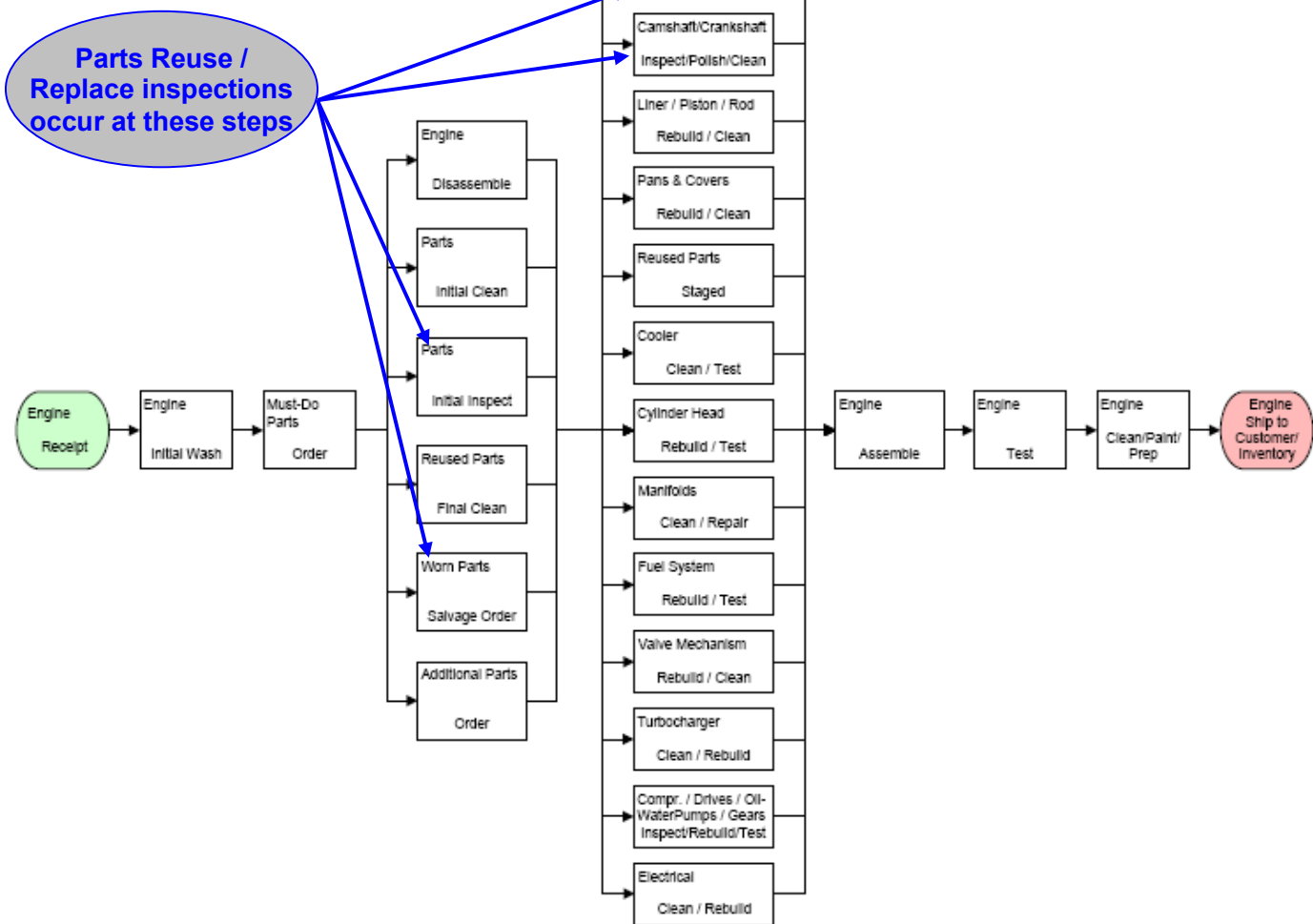
2.0 Best Practice Description

Some dealers have been very successful in managing parts reuse by creating a specialist level position in the shop. This position has been assigned overall responsibility for the parts reuse decisions and standard practices.

Responsibilities include:

- Inspect all parts for reuse during the disassembly or parts cleaning steps of the rebuild process.
- Facilitate the ordering of parts being replaced or transported for salvage, early in the rebuild process, to avoid delay/interruption during assembly.
- Assist with warranty failure analysis.
- Establish shop standard reuse policies and practices.
- Create and maintain Required Bill of Materials (RBOM) and/or “Must Replace” parts kits for the most common component rebuilds.
 - Establish parts reuse performance analysis reporting for training program development.
- Measure program effectiveness through annual scrap analysis and “Re-do” analysis.
- Participate in CRC cost management initiatives.
- Conduct parts reuse, and salvage training.
- Assume lead-person responsibilities in disassembly and cleaning areas. Coach technicians in proper disassembly & assembly techniques

Component Rebuild Center (CRC) Engine Level I Process Map - Single Bay Assembly



3.0 Implementation Steps

- The individual must be well trained in reusability guidelines, applied failure analysis, and the use of inspection/measurement equipment or tooling.
 - The Reuse Specialist should be available for each shift of a multi-shift operation. Due to the importance of the position and the daily workload, a back-up person should also be available and qualified.
 - The Reuse Specialist would work primarily in the Disassembly or Parts Cleaning area in an “assembly line” rebuild process. For a “Bay Build” process, the reuse specialist would float from bay to bay throughout disassembly.
 - A reuse inspection workstation may be required with a computer terminal, good lighting, magnification equipment, crack detection and other inspection and measuring equipment. Close access to magnaflux, crankshaft and camshaft inspection benches also helpful.

4.0 Benefits

- Manage parts reuse decisions as part of a consistent process.
- Remove repair decisions from the hands of lesser-experienced people.
- Reduce rebuild costs and / or re-do risks.
- Impact CRC margins and profit.
- Improve rebuild quality and component life.
- Improve shop efficiency – (specifically if “Must Do” parts kits are developed).

Other examples of benefits, which dealers have reported in process changes made to standard CRC practices:

- Process change requiring all engine blocks to be resurfaced and stainless steel water insert installed.
- Process change requiring all connecting rods to be magnafluxed.
- Process change requiring the fork spread of camshaft followers to be measured prior to reuse.

5.0 Resources Required

- A qualified individual for Parts Reuse Specialist should have extensive mechanical or engineering experience.

6.0 Supporting Attachments / References

Component Reuse Training dates and contact:

<https://dealer.cat.com/en/ps/service/rpe/c/2017-training-schedule.html>

DPC course # 41658

Why Reuse and Salvage Parts (info and case study):

<https://channel1.videonitch.com/LearningPrograms/Pass/1640?CategoryID=0>

7.0 Related Best Practices

None applicable at this time.

8.0 Acknowledgements

This Best Practice was written by:

Dale Brehm
Caterpillar Global Mining
Six Sigma Black Belt – Product Support Group